

# The Race between Man and Machine

Automation, new tasks, wages, and factor shares

Repository 6 — Artificial Intelligence and Economic Modeling

Acemoglu & Restrepo (AER, 2018)

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# Research question and core answer

## Question

Does automation necessarily reduce wages and the labor share?

- Automation must first change *effective* task allocation:  $I$  is technological capability, while  $I^*$  is equilibrium automation.
- If automation reallocates tasks, the labor share falls through **displacement**.
- The wage is conditional because displacement competes with a positive **productivity effect**.
- New labor-intensive tasks create a **reinstatement effect**.

## Bottom line

The labor-share effect is one-sided under the relevant static conditions; the wage effect is not.

# Continuum of tasks and final output

Tasks are indexed on a continuum:

$$i \in [N - 1, N].$$

Each task produces an intermediate  $y(i)$ , and the final good aggregates all tasks with constant elasticity:

$$Y = \left( \int_{N-1}^N y(i)^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}}.$$

- A task is an activity required to produce final output.
- The interval has unit measure;  $N$  locates the frontier of task content.
- Task allocation is therefore an endogenous margin of aggregate production.

# Task technologies and comparative advantage

For tasks that machines can perform ( $i \leq I$ ), either factor may be used:

$$y(i) = [k(i) + \gamma(i)l(i)]^{1-\eta} q(i)^\eta.$$

For nonautomatable tasks ( $i > I$ ), labor is required:

$$y(i) = [\gamma(i)l(i)]^{1-\eta} q(i)^\eta.$$

- $\gamma(i)$  is increasing: labor has comparative advantage in higher-index tasks.
- Effective factor costs are  $R$  for capital and  $W/\gamma(i)$  for labor.
- An automatable task uses capital when  $R < W/\gamma(i)$ .

## Technological frontier $I$ versus effective frontier $I^*$

Define the cost frontier by

$$\frac{W}{R} = \gamma(\tilde{I}).$$

Equilibrium automation is

$$I^* = \min\{I, \tilde{I}\}.$$

### Technology constrained

$I < \tilde{I}$ , hence  $I^* = I$ . A rise in  $I$  moves the effective frontier and reassigns the marginal task to capital.

### Cost constrained

$\tilde{I} < I$ , hence  $I^* = \tilde{I}$ . A rise in  $I$  expands feasibility but does not change actual task assignment.

Technological possibility is not the same as effective automation.

## Displacement effect

Consider a static automation-only change with fixed  $K$ :

$$dl > 0, \quad dN = 0, \quad I^* = I < \tilde{I}.$$

- ① Newly automatable marginal tasks move from labor to capital.
- ② Labor is compressed into a smaller set of tasks.
- ③ Diminishing returns within the remaining labor tasks reduce relative labor demand.

Proposition 2 gives

$$\frac{d \ln(W/R)}{dl} = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} < 0.$$

Thus effective automation lowers  $W/R$ , the labor share, and employment under the paper's labor-supply specification.

## Productivity effect and the wage

Automation also replaces expensive effective labor with cheaper capital:

$$d \ln Y|_{K,L} > 0.$$

For the technology-constrained allocation, Proposition 3 decomposes the wage:

$$d \ln W = d \ln Y|_{K,L} + (1 - s_L) \left[ \frac{\Lambda_N}{\hat{\sigma} + \varepsilon_L} dN - \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} dI \right].$$

With  $dN = 0$ :

$$\underbrace{d \ln W}_{\text{ambiguous}} = \underbrace{d \ln Y|_{K,L}}_{\text{productivity (+)}} - \underbrace{(1 - s_L) \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} dI}_{\text{displacement (-)}}.$$

Output can rise while the wage falls.

## When can automation raise the wage?

Under AER Assumptions 1–3, Proposition 3 establishes a finite threshold  $\tilde{K}$ :

$$K > \tilde{K} \implies \frac{d \ln W}{d l} > 0, \quad K < \tilde{K} \implies \frac{d \ln W}{d l} < 0.$$

- At low  $K$ , the cost-saving gap at the marginal task is small: displacement dominates.
- At high  $K$ , capital is abundant and the productivity gain can dominate.
- This concerns the *absolute wage*; it does not reverse the fall in  $W/R$  or the labor share.

### Source discipline

Published AER version, Proposition 3, pp. 1501–1503.

## Labor share and employment

Why can the wage rise while the labor share falls?

$$s_L = \frac{WL}{RK + WL}, \quad \frac{d \ln(W/R)}{dI} < 0.$$

- Productivity can increase the level of  $W$ .
- Displacement changes relative factor demand in favor of capital.
- Hence  $W$  and  $W/R$  need not move together.
- Under equation (4)'s labor-supply preferences, automation raises output per worker more than the wage, so employment falls.

The capital share rises as the complementary net-output factor share.

## New tasks and reinstatement

A higher  $N$  introduces a new high-index task where labor has comparative advantage. Under Assumption 3,

$$R > \frac{W}{\gamma(N)},$$

so the new task is adopted with labor.

| Change     | Output | Wage | Employment | Labor share | $W/R$ |
|------------|--------|------|------------|-------------|-------|
| Higher $N$ | +      | +    | +          | +           | +     |

This is **reinstatement**: labor returns to a wider range of activities. It is not merely more productivity; it changes the task content of production in labor's favor.

# Capital adjustment and endogenous technology

## Long-run capital adjustment: Proposition 5

On the relevant interior balanced-growth path,

$$R = \rho + \delta + \theta g.$$

Capital adjustment passes productivity gains toward the relatively inelastic factor. Long-run automation raises the wage but still lowers employment and the labor share.

## Direction of innovation: Proposition 6 and Corollary 2

A temporary automation imbalance can make further automation less attractive and new-task creation more attractive. A permanent shift toward automation instead leads to a new BGP with lower task content  $n = N - I$ , employment, and labor share.

## Does automation necessarily reduce wages and labor share?

| Experiment                               | Wage                                            | Labor share        |
|------------------------------------------|-------------------------------------------------|--------------------|
| $I$ rises but $I^*$ does not             | No marginal effect                              | No marginal effect |
| Effective automation, fixed $K$          | Conditional on productivity versus displacement | Falls              |
| Effective automation, $K$ adjusts on BGP | Rises under Proposition 5 conditions            | Falls              |
| New tasks                                | Rises                                           | Rises              |

### Rigorous answer

No. Effective automation reduces the labor share under the relevant conditions, but its wage effect depends on productivity, displacement, the capital stock, and the adjustment horizon.

## Where I did not believe the AI

AI initially wrote AER equation (5) as effective task cost divided by  $1 - \eta$ .

Why doubt it? That expression did not match the task technology or the displayed source equation.

AER says

$$p(i) = \min\{R, W/\gamma(i)\}^{1-\eta} \quad \text{for } i \leq I.$$

Correct interpretation The positive power preserves the cost ranking, so the allocation frontier is unchanged, but division would give wrong task-price levels.

Source: published AER, equation (5), p. 1496. This was a real transcription error, not a manufactured disagreement.

## Hand derivation: wage sign under automation

Start from Proposition 3, set  $dN = 0$ , and write  $d \ln Y|_{K,L} = P_I dl$ :

$$d \ln W = \left[ P_I - (1 - s_L) \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} \right] dl.$$

Because  $dl > 0$ ,

$$\frac{d \ln W}{dl} \geq 0 \iff P_I \geq (1 - s_L) \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L}.$$

Proposition 3 translates this comparison into

$$K > \tilde{K} \Rightarrow \frac{d \ln W}{dl} > 0, \quad K < \tilde{K} \Rightarrow \frac{d \ln W}{dl} < 0.$$

Yet Proposition 2 implies

$$\frac{d \ln(W/R)}{dl} = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} < 0.$$

### Manual evidence placeholder

Insert the real handwritten copy of `hand/derivation-guide.md` here after it is supplied. No synthetic handwriting has been generated.

# Lean formalization: the narrow claim

## Paper source

NBER Working Paper 22252, revised June 2017, Section 2.2, p. 8, equation (6) and the threshold discussion immediately following it.

$$\frac{w}{r} = \gamma(c), \quad e = \min\{a, c\}, \quad i < e.$$

Lean proves the implication

$$i < \min\{a, c\} \implies i \leq a \wedge r < \frac{w}{\gamma(i)}.$$

| Lean symbol               | Economic meaning                      |
|---------------------------|---------------------------------------|
| $a$ / automationThreshold | technological frontier                |
| $c$ / costThreshold       | cost frontier                         |
| $i$                       | task index                            |
| $w, r, \gamma$            | wage, rental rate, labor productivity |

# Lean proof, assumptions, and boundary

## Core proof logic (identifiers romanized for the slide)

```
have hiCost : i < costThreshold :=  
  lt_of_lt_of_le hi (min_le_right _ _)  
have hgammaLt : gamma i < w / r := by  
  rw [hthreshold]  
  exact hgamma hiCost  
have hmul : gamma i * r < w :=  
  (lt_div_iff0 hr).mp hgammaLt  
apply (lt_div_iff0 (hgammaPos i)).mpr  
simpa [mul_comm] using hmul
```

## Assumptions

- $\gamma$  strictly increasing and positive;
- $w > 0, r > 0$ ;
- $w/r = \gamma(c)$ ;
- $i < \min\{a, c\}$ .

**Verified:** a task below both frontiers is feasible for capital and capital is strictly cheaper.

**Not verified:** a full proposition, equilibrium, wage effects, shares, dynamics, or balanced growth.

## Actual result

Focused build passed; `paper_contribution.py check AR18RaceManMachine -fast` passed. Semantic closeout remains pending in the preserved audit scaffolding.

# Conclusion

- ➊ Only a change in  $I^*$ , not capability  $I$  alone, reallocates tasks.
- ➋ Effective automation creates displacement and productivity together.
- ➌ The labor share falls, but the wage can rise or fall at fixed capital;  $\tilde{K}$  separates the cases.
- ➍ New labor-intensive tasks reinstate labor and work in the opposite direction.
- ➎ Lean verifies one source-connected threshold implication—useful, but deliberately much narrower than the paper's economic propositions.

## Answer

Automation does not necessarily reduce both wages and the labor share.